BIOLOGY

1) Which of the following is a correct match regarding size/length ?

(1) Mycoplasma — 0.3 µm
(2) Viruses — 0.02 - 0.2 µm
(3) Human RBC — 7 µm
(4) All are correct

2) In plant cells, the vacuoles can occupy upto _____ % of the volume of the cell.

(1) 30
(2) 20
(3) 10
(4) 90

3) Which one statement is correct for cell membrane ?

(1) It is selectively permeable to molecules present on either side of it
(2) It allows neutral solutes to pass through only by carrier proteins
(3) Na^+/K^+ pump performs transport according to concentration gradient
(4) Water movement occurs by active transport

4) Which of the following is incorrect for Rudolf Virchow?

(1) He gave the statement 'Omnis cellula e cellula'
(2) He modified the hypothesis of Schleiden and Schwann to give cell theory a final shape
(3) According to him, all cells arise from pre-existing cells
(4) He proposed the cell theory but this theory did not explain how new cells are formed

5) Polyribosomes or polysomes represent:

(1) Microbodies like glyoxysomes
(2) Aggregation of m-RNA
(3) Aggregation of rRNA
(4) Several ribosomes attached to a single mRNA forming a chain

6) Which of the following organelles is present in plants but not in animals ?

(1) Ribosome
(2) Lysosome
(3) Peroxisome

(4) Plastid

7) How many of the following statements are correct ?
(a) All cells arise from pre-existing cells.
(b) Ribosomes and centriole are non-membrane bound organelles found in all cells eukaryotic as well as prokaryotic.
(c) The ribosomes of chloroplasts are smaller than the eukaryotic cytoplasmic ribosomes.
(d) In human beings, the membrane of the erythrocyte has approximately 52 percent lipids.
Options :

(1) Four
(2) One
(3) Three
(4) Two

8) Select the wrong match :

(1) Aleuroplast – Protein storage
(2) Elaioplast – Fat/oil storage
(3) Amyloplast – Starch storage
(4) Chromoplast – Chlorophyll storage

9) Cell wall :

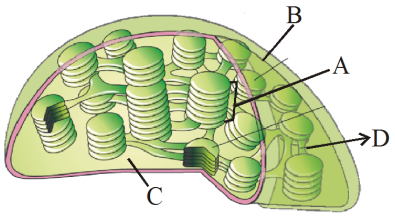
(1) is a living structure
(2) provides a free passage for undesirable macromolecules
(3) of algal cell is made of cellulose, galactans, mannans and calcium carbonate
(4) has middle lamella, mainly made of magnesium pectate

10) Which of the following is true for endoplasmic reticulum?

(1) It is involved in synthesis of lipid
(2) It is not included in endomembrane system
(3) It is found in some prokaryotes only
(4) It's main function is packaging of glycolipids

11) Which of the following is true for Mitochondria :

(1) Is bound by a single membrane
(2) Posses cristae which are extensions of its outer membrane
(3) Are the sites of aerobic respiration
(4) Is found in both eukaryotic and prokaryotic cells



12) Identify structure A,B,C and D is given below diagram :-

	A	B	C	D
(1)	Granum	Stroma lamella	Outer membrane	Stroma
(2)	Granum	Outer membrane	Stroma lamella	Stroma
(3)	Granum	Outer membrane	Stroma	Stroma lamella
(4)	Stroma lamella	Stroma	Outer membrane	Thylakoid

(1) 1

(2) 2

(3) 3

(4) 4

13) Match the following :

	Chromosomes		Position of centromere
(A)	Metacentric	(a)	At the end
(B)	Acrocentric	(b)	Slightly away from the middle
(C)	Telocentric	(c)	At the middle
(D)	Submetacentric	(d)	Almost near the end

Choose the correct match :-

(1) A-a, B-b, C-c, D-d

(2) A-c, B-d, C-a, D-b

(3) A-d, B-c, C-b, D-a

(4) A-b, B-a, C-d, D-c

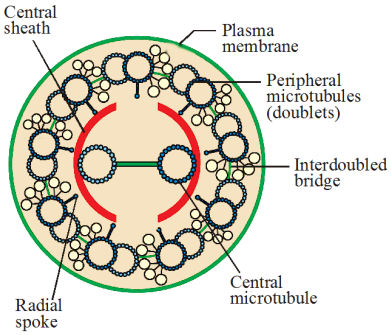
14) The ribosomes of eukaryotic cytoplasm, mitochondria and plastids are respectively :-

(1) 80S, 70S and 70S

(2) 70S, 70S and 80S

(3) 50S, 30S and 70S

(4) 70S, 80S and 70S



15) Which of the following is correct for the diagram given below?

(1) It represents prokaryotic flagella
(2) It possess '9 + 0' microtubular organisation
(3) Both (1) and (2)
(4) It represents hair-like outgrowths of cell membrane in eukaryotic ciliates and flagellates

16) Select the odd one out with respect to endo-membrane system :-

(1) Vacuole
(2) Golgi apparatus and Endoplasmic Reticulum
(3) Mitochondria and chloroplast
(4) Lysosome

17) Which of the following possess DNA ?

(1) Mitochondria
(2) Chloroplast
(3) Golgi apparatus
(4) Both (1) and (2)

18) Which of the following is/are non-membranous cell organelle(s)/structure(s)?

(1) Ribosome
(2) Nucleolus
(3) Centriole
(4) All of the above

19) How many of the given statements are correct?
I. Plasmodesmata connect cytoplasm of neighbouring animal cells.
II. Totally buried proteins in membrane are integral proteins while partialy buried proteins are peripheral proteins.
III. The primary cell wall is capable of growth.
IV. The peroxisome is not included in the endomembrane system.
Options :

(1) One
(2) Two
(3) Three

(4) Four

20) Nucleus is absent in :

(1) Erythrocytes of many mammals
(2) Mature sieve tube
(3) Both (1) and (2)
(4) All mature cells

21) Three of the following statements regarding the cell organelles are correct while one is wrong. Which one is wrong ?

(1) RER is frequently observed in the cells actively involved in protein synthesis
(2) The golgi apparatus principally performs the function of packaging materials
(3) Matrix of mitochondrion possesses circular DNA molecule
(4) Chlorophyll pigments are present only in the leucoplast

22) The cells which are secretory in function have abundant :-

(1) Mitochondria
(2) Golgi bodies
(3) Plastid
(4) Vacuoles

23) Which one is correct for lysosome ?

(1) It is non-membranous
(2) It lacks proteases
(3) It works at the acidic pH
(4) It is abundantly present in prokaryotic cells

24) The ability to move within membrane is measured as its fluidity. Degree of fluidity can be best measured by :-

(1) Amount of peripheral proteins in membrane
(2) Amount of saturated fatty acids in membrane
(3) lateral movement of proteins in membrane
(4) Lateral movement of phospholipids in membrane

25) Due to absence of which of the following prokaryotic cell is differ from eukaryotic cell ?

(1) Cytoplasm
(2) Nuclear membrane
(3) Ribosome
(4) Cell wall

26) Multinucleated condition in a cell arises when :-

(1) Karyokinesis is not followed by cytokinesis
(2) Cytokinesis is not followed by karyokinesis
(3) DNA replication is not followed by Chromosome splitting
(4) Chromosome splitting is not followed by karyokinesis

27) Which statement is not related with prophase ?

(1) Duplication of chromosomes take place
(2) Chromosomal material becomes untangled.
(3) Centrioles begin to move toward opposite poles.
(4) Chromosomes are seen to be composed of two chromatids.

28) Which one is the correct sequence of cell cycle ?

(1) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$
(2) $S \rightarrow G_2 \rightarrow G_1 \rightarrow M$
(3) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
(4) $M \rightarrow G_2 \rightarrow G_1 \rightarrow S$

29) It is said that the one turn of cell cycle in human cells takes 24 hours. Which phase of the cycle do you think occupies the maximum part of the cell cycle ?

(1) Prophase
(2) Interphase
(3) Anaphase
(4) Metaphase

30) In onion root cell there are 16 chromosomes. Number of chromosomes and chromatids is given in following stages :-

	Stage	**Chromosome**	**Chromatid**
(A)	G_1 phase –	16	16
(B)	Prophase –	16	32
(C)	Anaphase –	32	64
(D)	Metaphase –	32	32

Choose the wrong match ?
(1) A, B
(2) C, D
(3) Only A
(4) Only D

31) In monocotyledonous plants, the primary root is short-lived and is replaced by a large number of roots. These roots originate from the base of the stem and constitute the :

(1) Prop roots
(2) Pneumatophores
(3) Stilt roots
(4) Fibrous roots

32) Stem :

(1) Develops from radicle
(2) Bears nodes, internodes and buds
(3) Is involved in conduction of water only
(4) Does not perform the function of vegetative propagation

33)

Which phyllotaxy is not related to its example ?

(1) Alternate – China rose
(2) Opposite – *Calotropis*
(3) Whorled – *Alstonia*
(4) Opposite – Mustard

34) Arrangement of veins and veinlets in leaf lamina is known as :

(1) Inflorescence
(2) Phyllotaxy
(3) Venation
(4) Aestivation

35) How many plant have opposite phyllotaxy ?
Guava, *Calotropis*, *Alstonia*, Mustard, Sunflower, China rose

(1) 5
(2) 6
(3) 3
(4) 2

36) **Assertion :** Growth of axis is limited in cymose inflorescence.
Reason : The main axis terminates in a flower in cymose type of florescence.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

37) The corolla in which one posterior petal is large and two anterior ones forming a keel which are covered by two lateral wings is known as.

(1) Cruciform
(2) Funnel shaped
(3) Papilionaceous
(4) Bell shaped

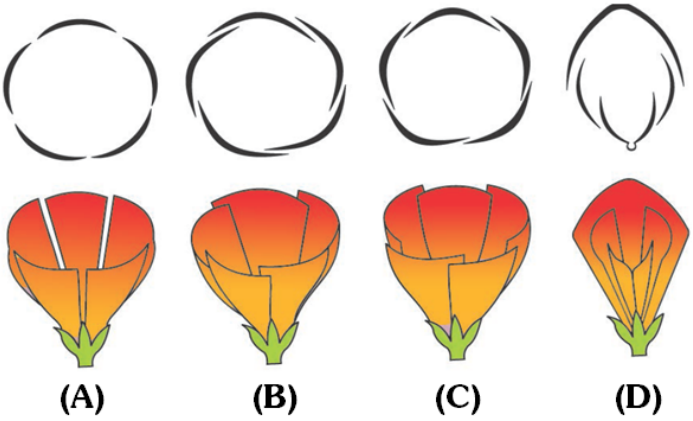
38) The flower which cannot be divided into two equal halves by any vertical plane is referred as :

(1) Actinomorphic flowers
(2) Zygomorphic flowers
(3) Radial symmetric flowers
(4) Asymmetric flowers

39) The condition in which stamens are attached with tapels is called :-

(1) Epipetalous
(2) Epiphyllous
(3) Gamotepalous
(4) Both 2 and 3

40) Identify the different type of aestivation shown in figures A, B, C and D and select the correct option :



(1) A–Valvate B–Twisted C–Imbricate D–Vexillary
(2) A–Valvate B– Imbricate C–Twisted D–Vexillary
(3) A–Vexillary B–Imbricate C–Twisted D–Valvate
(4) A–Twisted B– Imbricate C–Valvate D–Vexillary

41) When one margin of petals of a flower just touch one another at the margin with out overlapping, then such aestivation is known as:-

(1) Valvate

(2) Imbricate
(3) Vaxillary
(4) Twisted

42) Free central placentation is found in :

(1) *Dianthus*
(2) *Argemone*
(3) *Brassica*
(4) *Citrus*

43) **I–** When carpels are free, they are called __A__ as present in lotus and rose.
II– When carpel are fused, they are called __B__ as present in mustard and tomato.

(1) A–Syncarpous B–Apocarpous
(2) A–Apocarpous B–Syncarpous
(3) A–Syngensious B–Synandrous
(4) A–Syngensious B–Syncarpous

44) In which of the following fruit develops from monocarpellary gynoecium with superior ovary ?

(1) Mango
(2) Tomato
(3) Sun flower
(4) Guava

45) The aleurone layer in maize grain is rich in :

(1) Proteins
(2) Starch
(3) Lipids
(4) Auxins

46) One cotyledon is found in seeds/grains of:

(1) Wheat and Maize
(2) Gram and Pea
(3) Bean and Gram
(4) Groundnut and Pea

47) Given floral formula $\oplus \male\female K_{2+2}\, C_4\, A_{2+4}\, \underline{G}_{(2)}$ belongs to the family :

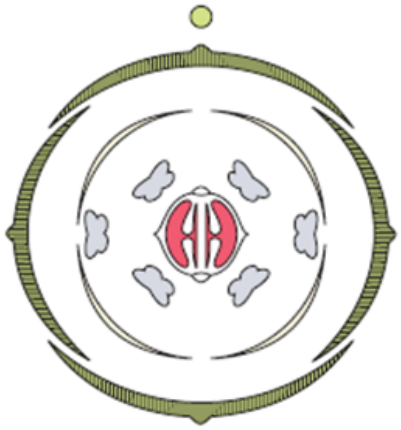
(1) Fabaceae
(2) Brassicaceae
(3) Liliaceae

(4) Solanaceae

48) Which floral symmetry is represented by the symbols ⊕ and %, respectively ?

(1) Zygomorphic and actinomorphic
(2) Actinomorphic and zygomorphic
(3) Asymmetric and zygomorphic
(4) Zygomorphic and Asymmetric

49) Which of the following feature not is shown by the given floral diagram ?



(1) Valvate aestivation
(2) 6, Polyandrous Stamens
(3) Parietal placentation
(4) Inflorescence usually cymose

50) Match the column I with column II and select the correct option :

	Column I		**Column II**
A	Bicarpellary syncarpous gynoecium and Superior ovary	i	$\underline{G}_{(2)}$
B	Calyx	ii	C
C	Corolla	iii	K
D	Androecium	iv	A

(1) A-i, B-ii, C-iii, D-iv
(2) A-i, B-iii, C-ii, D-iv
(3) A-ii, B-i, C-iii, D-iv
(4) A-iv, B-iii, C-ii, D-i

51) Bicarpellary, syncarpous gynoecium is found in members of family :-

(1) Liliaceae
(2) Fabaceae
(3) Solanaceae
(4) Malvaceae

52) Belladonna, which is used as medicine is obtained from the member of family :-

(1) Liliaceae
(2) Solanaceae
(3) Fabaceae
(4) Poaceae

53) Choose the correct floral formula with respect to Solanaceae family –

(1) $\oplus\ \male\female\ P_{(3+3)}\ A_{3+3}\ \underline{G}_{(3)}$

(2) $\oplus\ \male\female\ K_{(5)}\ C_{(5)}\ A_{5}\ \underline{G}_{(2)}$

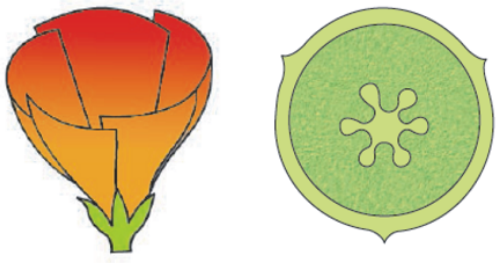
(3) $\%\ \male\female\ K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_{1}$

(4) $\%\ \male\female\ K_{(5)}\ C_{1+2+(2)}\ A_{5}\ \underline{G}_{(2)}$

54) Match the plants given in column-I with their uses given in column-II. Choose the option which gives the correct combination :-

Column-I		**Column-II**	
(A)	*Petunia*	(i)	As a spice
(B)	Tobacco	(ii)	As a medicine
(C)	Chilli	(iii)	As a fumigatory
(D)	*Atropa*	(iv)	As an ornamental plant

(1) A-ii B - iii C-iv D-i
(2) A-i B - iv C-iii D-ii
(3) A-iv B - iii C-i D-ii
(4) A-ii B -iv C-iii D-i



(A) (B)

55) Given figures A and B represents respectively :-

(1) Valvate aestivation, axile placentation
(2) Twisted aestivation, free central placentation
(3) Vexillary aestivation, Parietal placentation
(4) Valvate aestivation, Basal placentation

56) Match column-I (types of aestivation in petals) with column-II (related examples) and select the correct option –

Column-I (Type of aestivation in petals)		Column-II (Related examples)	
(a)	Valvate	(i)	Pea
(b)	Twisted	(ii)	*Cassia*
(c)	Imbricate	(iii)	Cotton
(d)	Vexillary	(iv)	*Calotropis*

(1) a-i, b-ii, c-iii, d-iv

(2) a-iii, b-iv, c-i, d-ii

(3) a-iv, b-iii, c-i, d-ii

(4) a-iv, b-iii, c-ii, d-i

57) Consider the following four statements (a-d) and select the option which includes all the correct ones only:-
(a) In Solanaceae family, flowers are zygomorphic
(b) In Solanaceae family, ten stamens are present in a flower
(c) In Solanaceae family, seeds are endospermic. (d) In Solanaceae family, axile placentation is present.

(1) Only statement (a) and (b)

(2) Only statement (b) and (c)

(3) Statement (a), (b), (c) and (d)

(4) Only statement (c) and (d)

58) Some examples are given below how many of them have both following characters ? (1) Axile placentation
(2) Adhesion of stamen Mustard, Sunflower, *Dianthus*, *Argemone*, Tomato, Potato, *Petunia*, *Sesbania*, *Trifolium*, Primrose.

(1) Three

(2) Four

(3) Seven

(4) Five

59) Which conditions are not found in Solanaceae family ?
(a) Oblique ovary and swollen placenta
(b) Fruit - Lomentum
(c) Gamosepalous
(d) Polypetalous
(e) Epiphyllous stamen
(f) Unilocular ovary and mono carpellary gynoecium
(g) Axile placentation
(h) Diadelphous stamen
(i) Endospermic seed

(1) a,c,g,i

(2) b,e,f,g,h

(3) b,d,e,f,h

(4) b,c,e,f,i

60) **Assertion :** Superior ovary is found in mustard and china rose.
Reason : In mustard and china rose gynoecium occupies the highest position while the other parts are situated below it.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

61) Which of the following is not a part of ground tissue system ?

(1) Pericycle and hypodermis

(2) Pith and pith rays

(3) Trichome and root hairs

(4) Cortex and Endodermis

62) Conjoint type of vascular bundles are common in :-

(1) Only stems

(2) Only leaves

(3) Only roots

(4) Both stems and leaves

63) Select the mismatched pair.

(1) Subsidiary cells - Parenchymatous

(2) Epidermial appendages - Trichomes and root hairs

(3) Cuticle - Present in roots

(4) Stomatal apparatus - Stomatal aperture, Guard cell and Subsidiary cell

64) When xylem and phloem within a vascular bundle are arranged in an alternate manner on different radii then it is known as-

(1) Radial

(2) Conjoint

(3) Bicollateral

(4) Open

65) Closed vascular bundle lack which of the following ?

(1) Xylem

(2) Phloem

(3) Cambium

(4) Both (1) and (2)

66) The type of vascular bundles present is dicot stem is :-

(1) Open, conjoint with endarch

(2) Open, conjoint with exarch

(3) Closed, conjoint with endarch

(4) Closed, conjoint with exarch

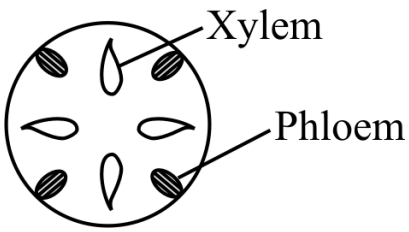
67) How many in the list given below are the part of ground tissue system.
Epidermis, Root hairs, Xylem, Phloem, Cortex, Pericycle, Pith.

(1) One

(2) Two

(3) Three

(4) Four



68) This type of vascular bundle are called ?

(1) Radial

(2) Collateral

(3) Endarch

(4) 1 and 3

69) Usually the guard cells are ___A___. The outer wall of guard cells away from the stomatal pore are ___B___ and inner walls (towards the stomatal pore) are ___C___. Select correct option for A, B & C.

(1) A – Bean shaped, B – thick, C – Less thickned

(2) A – Dumb-bell shaped, B – thin, C – Highly thickned

(3) A – Kidney shaped, B – thin, C – Highly thickned

(4) A – Dumb-bell shaped, B – thick, C – Less thickned

70) **Statement-I :** The root hairs are unicellular elongations of the epidermal cells and help absorb water and minerals from the soil.
Statement-II : The trichomes in the root system are usually multicellular and they may be branched or unbranched and soft or stiff.

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is true but Statement II is false

(3) Statement I is false but Statement II is true

(4) Both Statement I and Statement II are correct.

71) Which one is not true for ground tissue system.

(1) It consists of simple permanent tissue such as parenchyma, collenchyma & sclerenchyma
(2) It is largest tissue system.
(3) It helps in preventing water loss due to transpiration
(4) In leaves, mesophyll cells are included in ground tissue system

72)

Match the column-I with column-II & select correct option :-

Column-I		**Column-II**	
(a)	Dicot stem	(i)	Open vascular bundle
(b)	Monocot stem	(ii)	Closed vascular bundle
(c)	Root	(iii)	Radial vascular bundle
(d)	Stem and leaves	(iv)	Conjoint vascular bundle

Choose the correct answer from the options given below :
(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(4) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

73) Stomatal apparatus does not includes :-

(1) Stomatal aperture
(2) Guard cells
(3) Subsidiary cells
(4) Cuticle

74) In dicot leaves, the guard cells are :-

(1) Barrel shaped
(2) Kidney shaped
(3) Oval shaped
(4) Dumb-bell shaped

75) Select the incorrect statement :-

(1) Epidermal cells are parenchymatous.
(2) Cuticle prevents the loss of water in roots.
(3) The trichomes in the shoot system are usually multicellular.

(4) The root hairs are unicellular

76) **Statement-I :** Each stomata is usually composed of two bean shaped cell known as guard cells.
Statement-II : In dicotyledon stem cambium is present between xylem and phloem.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are incorrect.

77) 
Identify the given diagram and select correct option about diagram.

(1) Conjoint open vascular bundle
(2) Present in stem of sunflower
(3) Unable to form secondary xylem and phloem tissue
(4) Present in monocot roots

78) The inner most layer of the cortex is called as :-

(1) Hypodermis
(2) Endodermis
(3) Epidermis
(4) Pericycle

79) In monocot root which type of vascular bundles are present ?

(1) Collateral, conjoint and closed
(2) Collateral, conjoint and open
(3) Bicollateral, conjoint and open
(4) Radial

80) In dicot root, cells of which region show casparian strip ?

(1) Hypodermis
(2) Pith
(3) Endodermis
(4) Pericycle

81) In grasses :-

(1) Certain adaxial epidermal cells along the veins modify themselves into large empty colourless cells, these are called bulliform cells.

(2) When the bulliform cells in the leaves absorb water and become flacid and leaf surface area is reduced.

(3) When bulliform cells become turgid due to water stress and surface area is reduced.

(4) When bulliform cells in leaves absorb water and become turgid and leaf surface area is reduced.

82) **Statement-I :** There are usually more than six (polyarch) xylem bundles in the dicot root.
Statement-II : There are usually two to four xylem and phloem patches in monocot root.

(1) Both statements are correct.

(2) Both the statements are incorrect.

(3) Statement I is correct and statement II is incorrect.

(4) Statement I is incorrect and statement II is correct.

83) Which of the following is not associated with root ?

(1) Casparin strips on endodermis.

(2) Conjuctive tissue

(3) Radial vascular bundles

(4) Hypodermis

84) What is the position of conjuctive tissue in dicot root ?

(1) Above the phloem

(2) Between xylem and phloem

(3) In the center

(4) Between inter and intra fascicular cambium

85) In endodermis of dicot roots, deposition of _____ forms cabparian strips :

(1) Lignin

(2) Suberin

(3) Algin

(4) Resin

86) Which of the following forms hypodermis of monocot stem ?

(1) Chlorenchyma

(2) Collenchyma

(3) Sclerenchyma

(4) Parenchyma

87) A plant structure has a sclerenchymatous hypodermis, large number of scattered vascular bundles, each surrounded by sclerenchymatous bundle sheath. This structure will be :-

(1) Dicot stem
(2) Monocot stem
(3) Dicot root
(4) Monocot root

88) Many of the cells of _____protrude in the form of unicellular root hairs.

(1) Hypodermal
(2) Epiblema
(3) Endodermal
(4) Cortical

89) Which of the following is not a character of monocot stem ?

(1) Large number of scattered vascular bundles.
(2) Parenchymatous bundle sheath
(3) Closed vascular bundles.
(4) Presence of water containing cavities

90) Select the incorrect statement regarding dicotyledon stem :-

(1) Hypodermis of dicotyledon stem is made up of parenchyma.
(2) Multiple layers between epidermis and pericycle are present which are known as cortex.
(3) The cells of the endodermis are rich in starch grains.
(4) In pericycle, above the phloem semi-lunar patches of sclerenchyma are present.

ANSWER KEYS

BIOLOGY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**4**	**4**	**1**	**4**	**4**	**4**	**4**	**4**	**3**	**1**	**3**	**3**	**2**	**1**	**4**	**3**	**4**	**4**	**2**	**3**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**4**	**2**	**3**	**3**	**2**	**1**	**1**	**3**	**2**	**2**	**4**	**2**	**4**	**3**	**4**	**1**	**3**	**4**	**2**	**1**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**1**	**1**	**2**	**1**	**1**	**1**	**2**	**2**	**4**	**2**	**3**	**2**	**2**	**3**	**2**	**4**	**4**	**1**	**3**	**1**
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	**3**	**4**	**3**	**1**	**3**	**1**	**3**	**1**	**3**	**2**	**3**	**3**	**4**	**2**	**2**	**1**	**3**	**2**	**4**	**3**
Q.	81	82	83	84	85	86	87	88	89	90										
A.	**1**	**2**	**4**	**2**	**2**	**3**	**2**	**2**	**2**	**1**										

SOLUTIONS

BIOLOGY

1)

Explain Question : Which of the following is a correct match regarding size/length ?
Concept : This question is based on Cell Size and Shape
Explanation :
Mycoplasma — 0.3 µm → Correct
Mycoplasma are the smallest living cells, about 0.3 µm in size.
Viruses — 0.02 – 0.2 µm → Correct
Viruses range from 20 nm to 200 nm (0.02 – 0.2 µm).
Human RBC — 7 µm → Correct
Average diameter of human red blood cell = 7.0 – 7.5 µm.
Final answer : 4

2)

Explain the Question : The question asks about the maximum proportion of the plant cell volume that vacuoles can occupy.
Concept : This question based on Vacuoles
Explanation : Vacuoles are large membrane-bound organelles in plant cells, primarily filled with cell sap.
Typical estimates suggest up to 90% of the cell volume can be taken up by the vacuole.
Final Answer : 4 – 90

3)

Explain Question : Which one statement is correct for cell membrane ?
Concept : This question is based on Plasma Membrane: Transport across membrane
Explanation : Plasma membrane is selectively permeable, meaning it allows only certain substances to pass through.
Final answer : 1

4) **Explain Question:** The question asks to identify the incorrect statement about Rudolf Virchow and his contributions to cell theory.
Concept : This question is based on cell theory
Explanation : He proposed the cell theory but this theory did not explain how new cells are formed. incorrect
Rudolf Virchow (1855):

- Proposed “Omnis cellula e cellula” → every cell arises from a pre-existing cell.
- Modified Schleiden and Schwann’s ideas to finalize cell theory.

Final answer : 4

5)

NCERT XI, Eng. Pg # 129
NCERT XI, हिन्दी पेज # 129

6)

Explain the Question : The question asks which organelle is present in plant cells but absent in animal cells.
Concept : This question based on Plastids
Explanation : Plastids – Found only in plant cells (chloroplasts, chromoplasts, leucoplasts), involved in photosynthesis and storage
Final Answer : 4 – Plastid

7)

Explain Question: The question asks that how many of the given statements are correct.

Concept: Cell theory/ ribosomes

Solution:

The correct statements are:

(a) All cells arise from pre-existing cells.

(c) The ribosomes of chloroplasts are smaller than the eukaryotic cytoplasmic ribosomes.

So, the correct answer is : 4. (Two)

Final answer: 4.

8)

Explain the Question : The question asks to identify the incorrect match between types of plastids and their storage or pigment functions.
Concept : This question based on Plastids
Explanation : Chromoplast – contains carotenoids (yellow, orange, red pigments), not chlorophyll. Chlorophyll is in chloroplasts.
Final Answer : 4 – Chromoplast – Chlorophyll storage

9) NCERT XI, Eng. Pg. # 132
NCERT XI, हिन्दी पेज # 132, 133

10)

Question Explanation:

Function of endoplasmic reticulum.

Concept:

Endoplasmic Reticulum

Solution:

The endoplasmic reticulum (ER) is a network of membranes inside eukaryotic cells.
It has two parts: the smooth ER and the rough ER.
The smooth ER is responsible for synthesizing lipids, steroids, and hormones, while the rough ER modifies and transports proteins.
The ER is also part of the endomembrane system, and it is not found in prokaryotes.

Final Answer:

It is involved in synthesis of lipid.

11) **Explain the Question :** The question asks which statement correctly describes the "powerhouse of the cell
Concept : This question based on Mitochondria
Explanation : Mitochondria are double-membrane organelles.
The inner membrane forms cristae, which increases the surface area for aerobic respiration.
They are the site of aerobic respiration and ATP production.
Mitochondria are present in eukaryotic cells only, not in prokaryotes.
Final Answer : 3 – Are the sites of aerobic respiration

12)

NCERT-XI, Eng. Pg. # 136, [Fig. 8.8]
NCERT-XI, हिन्दी पेज # 136, [चित्र 8.8]

13)

NCERT-XI, Eng. Pg. # 139
NCERT-XI, हिन्दी पेज # 139

14)

NCERT XI, Eng. Pg # 135, 136
NCERT XI, हिन्दी पेज # 135, 136

15) **Explain the Question :** The question asks about the structure shown in a diagram, likely a flagellum or cilium, and its characteristics.
Concept : This question based on flagellum or cilium.
Explanation :
Hair-like outgrowths of eukaryotic cells are cilia or flagella, composed of microtubules.
Final Answer : 4 – It represents hair-like outgrowths of cell membrane in eukaryotic ciliates and flagellates

16)

NCERT XI, Eng. Pg # 133
NCERT XI, हिन्दी पेज # 133

17)

NCERT XI, Eng. Pg # 135, 136
NCERT XI, हिन्दी पेज # 135, 136

18)

Explain Question : The question asks which among the given options are non-membranous cell organelles/structures.
Concept : This question is based on Ribosome , Nucleolus ,Centriole
Explanation :

A. Ribosome – Non-membranous, made of rRNA and proteins, site of protein synthesis.

B. Nucleolus – Non-membranous, present inside the nucleus, site of rRNA synthesis and ribosome assembly.

C. Centriole – Non-membranous, composed of microtubules (9+0 arrangement), helps in spindle formation.

Final answer : 4

19) III and IV statements are correct.
NCERT-XI Eng. Pg. # 131, 132, 133
कथन III एवं IV सही है।
NCERT-XI हिन्दी पेज-131, 132, 133

20)

Explain Question : The question asks in which of the given cases the nucleus is absent
Concept : This question is based on nucleus
Explanation :

A. Erythrocytes (RBCs) of many mammals lose their nucleus at maturity to increase space for hemoglobin and improve oxygen transport.

B. Mature sieve tube elements (phloem tissue) also lack a nucleus to allow efficient translocation of food but depend on companion cells for functioning.

Final answer : (3)

21)

Explain Question : Question asking about identify the wrong statement among the four regarding cell organelles.
Concept : This question is based on plastid, Mitochondria, E.R. and Golgi body
Solution : RER for protein synthesis → correct.
Golgi apparatus → packaging → correct.
Mitochondrial matrix has circular DNA → correct.
Chlorophyll in leucoplast → wrong, it is in chloroplasts.
Final answer : 4

22)

NCERT XI, Eng. Pg # 134
NCERT XI, हिन्दी पेज # 134

23)

Explain Question : The question asks which statement about lysosomes is correct. Lysosomes are cell organelles containing hydrolytic enzymes.
Concept : This question is based on Lysosomes
Explanation :
Lysosomes are single-membrane bound organelles, not non-membranous.
They contain hydrolytic enzymes, including proteases, lipases, nucleases.
These enzymes are active at an acidic pH (~5) maintained by proton pumps
Final answer : 3

24)

NCERT XI, Eng. Pg. # 131, 132
NCERT XI, हिन्दी पेज # 131

25)

NCERT XI Eng. Pg. # 126
NCERT XI हिन्दी पेज # 126

26) NCERT XI, Eng. Pg.-166
NCERT XI, हिन्दी पेज-167

27)

Question explanation : The question is asking which statement does not describe events of prophase.

Concept: Prophase.

Solution: Chromosome duplication occurs in S-phase, not in prophase. The other statements describe prophase accurately: chromosomes condense, become clearer, and centrioles move to opposite poles.

Final answer: Option 1

28) NCERT-XI, Eng. Pg. # 163
NCERT-XI, हिन्दी पेज # 163

29)

NCERT-XI, Eng. Pg. # 163
NCERT-XI, हिन्दी पेज # 163

30)

NCERT XI, Eng. Pg.# 164, 165, 166 (Topic # 10.2)
NCERT XI, हिन्दी पेज.# 164, 165, 166 (टॉपिक # 10.2)

31) NCERT Pg # 65 (5.1)

33)

NCERT XI, Pg. No. 71

34)

Correct Answer: (3)

36) NCERT XI Page No. # 72

37) NCERT XI, Pg # 79

38) NCERT XI Pg.# 72

39) NCERT XI, Pg # 75

40) NCERT XI Pg.# 74

41) NCERT XI, Pg # 74

42) NCERT XI Pg.# 75

43) NCERT XI Pg.# 75

44) **Explain the Question :** The question asks which fruit develops from a monocarpellary gynoecium with a superior ovary.
Concept : This question is based on morphology of flowering plants-fruit.
Solution :
Drupe develops from a monocarpellary, superior ovary (e.g. mango).
Pepo develops from an inferior ovary (Cucurbitaceae).
Pome is a false fruit and develops from an inferior ovary (apple).
Only drupe arises from a monocarpellary gynoecium with superior ovary.
Final Answer : (1)

45) NCERT XI Pg.# 77

46) NCERT XI Pg.# 77

47) **Explain Question –**
The question asks which plant family corresponds to the floral formula:
⊕ □ K_{2+2} C_4 A_{2+2} $G_{(2)}$

Concept – Brassicaceae family

Solution –
K 2+2 → Sepals in two whorls (outer 2 + inner 2)
C 4 → Four petals
A 2+4 → Tetradynamous stamens (2 short + 4 long)
G (2) → Bicarpellary, syncarpous, superior ovary
These features are diagnostic of the Brassicaceae family

Final Answer – Option 2

48) NCERT XI Pg.# 78

49) NCERT XI Pg.# 78

50) NCERT XI Pg.# 78

51) NCERT-XIth, Pg. # 80

52) NCERT-XIth, Pg.# 80

53) NCERT-XIth, Pg.# 80

54) NCERT-XIth, Pg.#.#80

55) NCERT XI, Pg # 74,75

56) NCERT XIth Pg. # 74

57)

Explain Question : Question asking about floral characters of the Solanaceae family of angiosperms.
Concept : This question is based on Solanaceae family
Solution :
(a) In Solanaceae family, flowers are zygomorphic- **Incorrect**
In Solanaceae family, flowers are actinomorphic
(b) In Solanaceae family, ten stamens are present in a flower- **Incorrect**
In Solanaceae family, five stamens are present in a flower
(c) In Solanaceae family, seeds are endospermic.-**correct**
(d) In Solanaceae family, axile placentation is present.-**correct**
Final Answer : option (4)

58)

NCERT Pg. # 69

59)

Explain Question : Question asks which listed conditions are NOT found in the family Solanaceae.
Concept : Morphology of Flowering Plants- Floral characters of Solanaceae
Solution :
(a) Oblique ovary with swollen placenta -Found in Solanaceae
(b) Fruit – Lomentum - Fabaceae feature, not Solanaceae
(c) Gamosepalous - Found in Solanaceae
(d) Polypetalous - Solanaceae is gamopetalous
(e) Epiphyllous stamen -Character of Liliaceae
(f) Unilocular ovary and monocarpellary gynoecium - Solanaceae is bicarpellary
(g) Axile placentation - Found in Solanaceae
(h) Diadelphous stamen - Fabaceae feature
(i) Endospermic seed -Found in Solanaceae
Thus, conditions not found are b, d, e, f, h.
Final answer - Option (3)

60)

Explain Question: The question is asking to check whether the Assertion and Reason are

true, and whether the Reason correctly explains the Assertion.

Concept: Position of floral parts on thalamus

Solution:

Assertion: Superior ovary is found in mustard and china rose → True
(Both have hypogynous flowers, so the ovary is superior.)

Reason: In mustard and china rose, gynoecium occupies the highest position while other parts are situated below it → True

This is exactly the definition of a hypogynous flower, which explains why the ovary is superior.

So, the Reason correctly explains the Assertion.

Final answer: Option 1

61) NCERT, Pg. # 73

62) NCERT, Pg. # 73

63) NCERT Page No. 72

64) NCERT Pg. # 73

65) NCERT, Pg. # 73

66) NCERT, Pg. # 73, 75

67)

NCERT-XI, Pg. # 72
Cortex, Pericycle, Pith

68)

Radial

69) NCERT, Pg. # 89

70) **Question Explanation:** Evaluate statements on plant epidermal structures.
Concept: Plant Tissue Anatomy.
Solution: Statement I is **correct** because **root hairs** are indeed unicellular extensions of epidermal cells and their primary function is to absorb water and minerals. Statement II is

incorrect because while trichomes can be branched, unbranched, soft, or stiff, they are present on the stem and leaves, not typically in the root system. The term for root epidermal outgrowths is root hairs.
Final Answer: Statement I is true but Statement II is false.

71) NCERT Pg # 89

72) NCERT-XI, Pg. # 90

73) NCERT Pg. # 72

74)

NCERT, Pg. # 72

75) NCERT, Pg. # 71-72

76) NCERT Pg. # 72, 73

77) NCERT-XI Pg#73

78)

NCERT, Pg. # 73

79) NCERT Pg # 73, 74

80) NCERT Pg # 73, 74

81) NCERT Pg # 73, 74

82) NCERT Pg # 73, 74

83) NCERT Pg. # 74

84) NCERT, Pg. # 74

85)

Explanation : Question is asking about in endodermis of dicot roots, deposition of which substance in the forms of casparian strips.

Concept: Anatomy of dicot and monocot root.
Solution: In dicot root the tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of casparian strips.
Final answer: Suberin.

86)

NCERT-XI, Pg. # 76

87) NCERT Pg. # 76

88) NCERT Pg. # 72

89)

Explanation : Question is asking which is not a character of monocot stem

Concept: Anatomy of monocotyledonous stem

Solution:

(2)Parenchymatous bundle sheath-not true for monocot stem

sclerenchymatous bundle sheath present in monocot stem

Final answer: Parenchymatous bundle sheath

90) NCERT, Pg. # 75-76